SpaceX Falcon 9 First-Stage Landing Data Science Capstone

Predicting launch outcomes with data science and machine learning

Your Name | IBM Data Science Professional Certificate

Date: October 21, 2026

GitHub: https://github.com/YOUR-USERNAME/ibm-ds-capstone-spacex

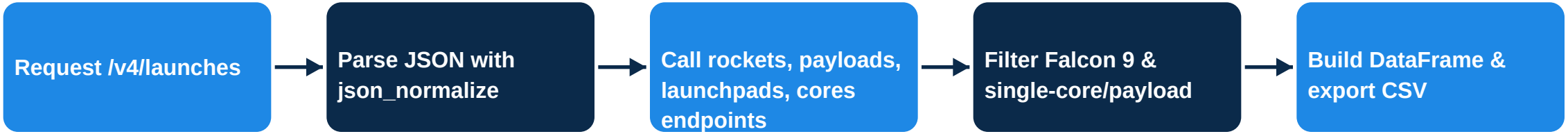
Executive Summary

- **Problem:** Can we predict whether a Falcon 9 first stage will land successfully? Reuse drives launch cost (~$62M vs. $165M+).
- **Methods:** Data collected via the SpaceX REST API and Wikipedia web scraping; cleaned with Pandas; explored with Matplotlib/Seaborn and SQL (SQLite); mapped with Folium; dashboarded with Plotly Dash.
- **Modeling:** Logistic Regression, SVM, Decision Tree and KNN tuned with GridSearchCV on standardized features.
- **Key results:** Success rate rose from 0% (2010) to ~84% (2020); orbits ES-L1, GEO, HEO, SSO had 100% success; KSC LC-39A had the highest success ratio; all models reached 83.3% test accuracy, with the Decision Tree best in cross-validation (88.9%).

Introduction

- **Background:** SpaceX advertises Falcon 9 launches at ~$62 million; competitors charge $165 million+. Savings come from reusing the first stage.
- **Business context:** If we can predict first-stage landing, we can estimate the cost of a launch — valuable to an alternate company bidding against SpaceX.
- **Problem statement:** Using public launch data, determine the factors (payload, orbit, launch site, booster version, reuse) that drive landing success and build a classifier to predict it.
- **Questions:** Which sites and orbits are most successful? How does payload mass matter? Has success improved over time? Which model predicts best?

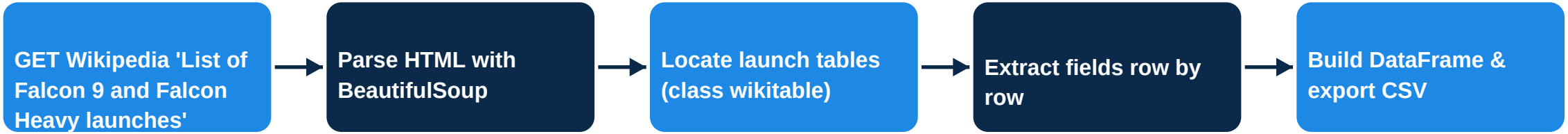
Data Collection – SpaceX API



- Key phrases: **requests.get()**, **.json()**, helper functions getBoosterVersion / getLaunchSite / getPayloadData / getCoreData
- Result: 90 Falcon 9 launches, 17 features (FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Reused, Legs, GridFins, Block, ReusedCount, Serial…)
- Missing PayloadMass values replaced with the column mean

Notebook: https://github.com/YOUR-USERNAME/ibm-ds-capstone-spacex/blob/main/01-spacex-data-collection-api.ipynb

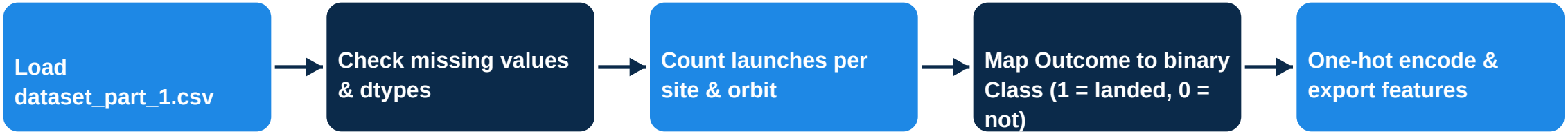
Data Collection – Web Scraping



- Key phrases: **requests.get()**, **BeautifulSoup**, **find_all('table')**, helper functions date_time / booster_version / landing_status
- Extracted: Flight No., Launch Site, Payload, Payload Mass, Orbit, Customer, Launch Outcome, Version Booster, Booster Landing, Date, Time
- Purpose: cross-check and enrich the API data with historical records

Notebook: https://github.com/YOUR-USERNAME/ibm-ds-capstone-spacex/blob/main/02-webscraping.ipynb

Data Wrangling Methodology



- Identified landing outcomes: True Ocean, True RTLS, True ASDS (success); False Ocean/RTLS/ASDS, None None, None ASDS (failure)
- Landing success rate for the dataset: **66.7%**
- Final feature set: FlightNumber, PayloadMass, Orbit, LaunchSite, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial

Notebook: https://github.com/YOUR-USERNAME/ibm-ds-capstone-spacex/blob/main/03-data-wrangling.ipynb

EDA with Data Visualization

- **Scatter plots** (Flight Number vs Launch Site; Payload vs Launch Site; Flight Number vs Orbit; Payload vs Orbit) — show how site, orbit, experience and payload relate to landing outcome.
- **Bar chart** (Success rate by Orbit) — compares orbit types and highlights where landings are most reliable.
- **Line chart** (Yearly success trend) — shows learning and improvement over time.
- **Feature engineering:** one-hot encoding of Orbit, LaunchSite, LandingPad, Serial; features cast to float64 for modeling.

Notebook: https://github.com/YOUR-USERNAME/ibm-ds-capstone-spacex/blob/main/05-eda-dataviz.ipynb

EDA with SQL

- Loaded the SpaceX table into an SQLite database via **%sql** magic and pandas.

- Queries performed: unique launch sites; 5 records where site starts with 'CCA'; total payload for NASA (CRS); average payload for booster F9 v1.1; date of first successful ground-pad landing; boosters with drone-ship success and payload 4,000–6,000 kg; counts of mission outcomes; boosters carrying max payload; 2015 failed drone-ship launches by month; ranking of landing outcomes between 2010-06-04 and 2017-03-20.

Notebook: https://github.com/YOUR-USERNAME/ibm-ds-capstone-spacex/blob/main/04-eda-sql.ipynb

Interactive Visual Analytics

Folium Map: launch site markers, circles, success/failure color-coded clusters, distance lines to coast, railway, highway, city

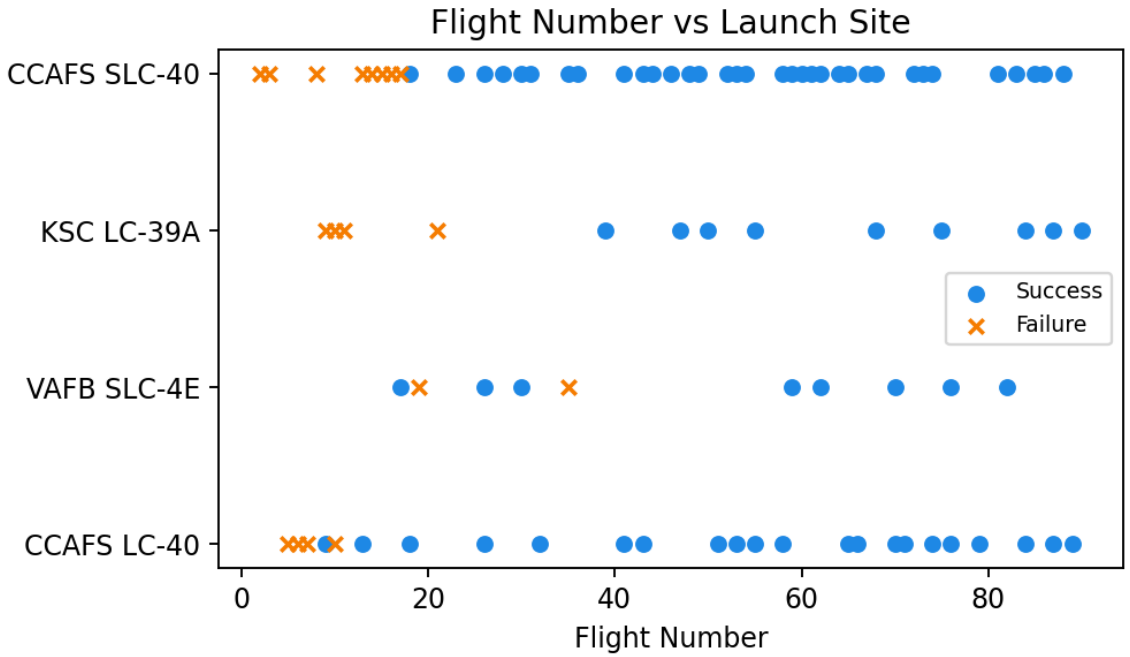
Plotly Dash App: site drop-down, success pie chart, payload range slider, payload vs outcome scatter plot

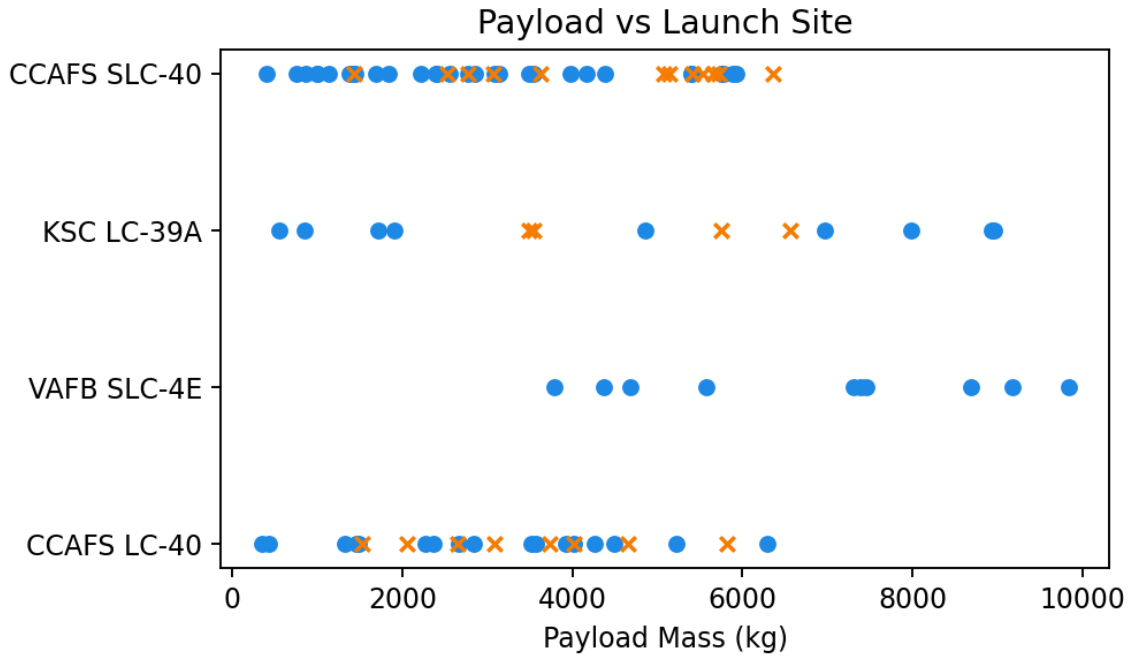
- Folium answers: where are sites located and how close are they to infrastructure?
- Dash answers: which site is most successful, and which payload ranges and boosters succeed most?

Folium: https://github.com/YOUR-USERNAME/ibm-ds-capstone-spacex/blob/main/06-folium-launch-sites.ipynb

Dash: https://github.com/YOUR-USERNAME/ibm-ds-capstone-spacex/blob/main/spacex_dash_app.py

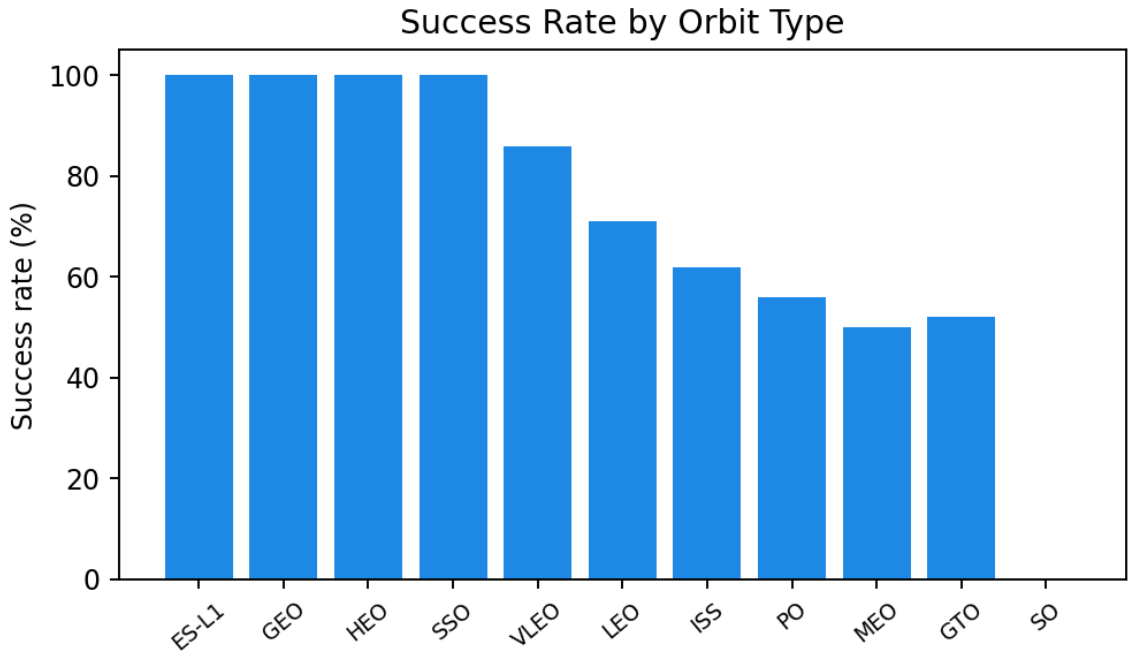
EDA Results: Scatter Plots

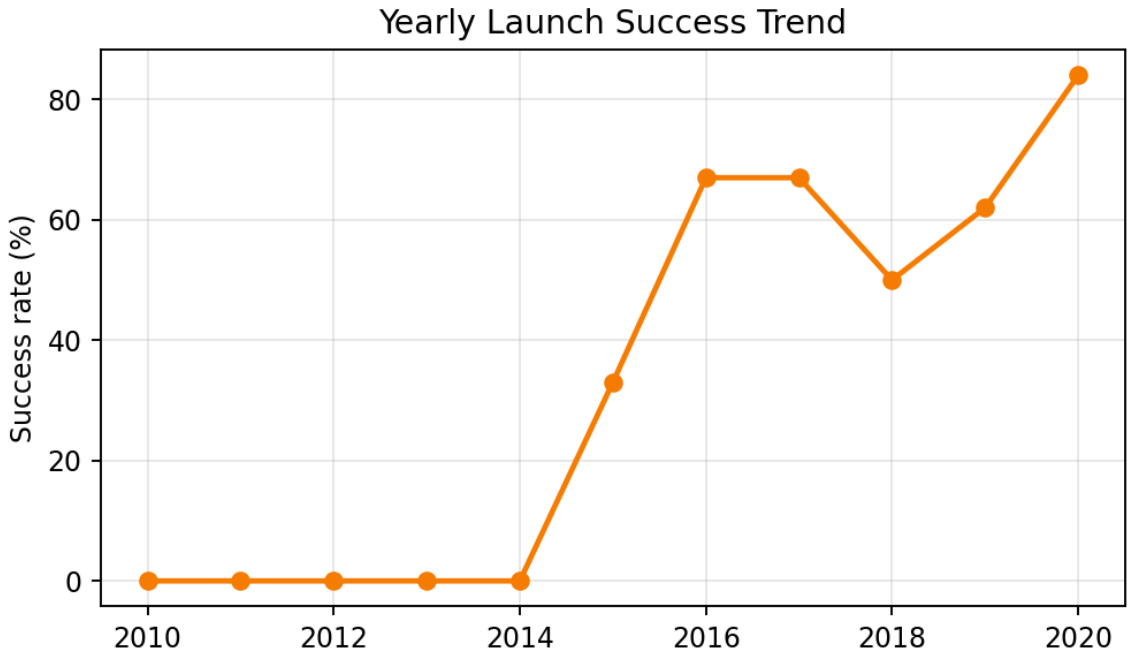




Later flights succeed more often; CCAFS SLC-40 has the most launches; VAFB has no heavy payloads above ~10,000 kg.

EDA Results: Bar Chart & Yearly Trend





ES-L1, GEO, HEO, SSO reached 100% success; success rate climbed steadily from 2013 and reached about 84% by 2020.

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EDA with SQL Results: Sites, Payloads, Success

Query	Result
Unique launch sites	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
Total payload (NASA CRS)	45,596 kg
Avg payload (F9 v1.1)	2,928 kg
First successful ground-pad landing	2015-12-22
Drone-ship success, 4,000-6,000 kg	F9 FT B1022, B1026, B1021.2, B1031.2
Mission outcomes	100 successes, 1 failure in flight, 1 partial

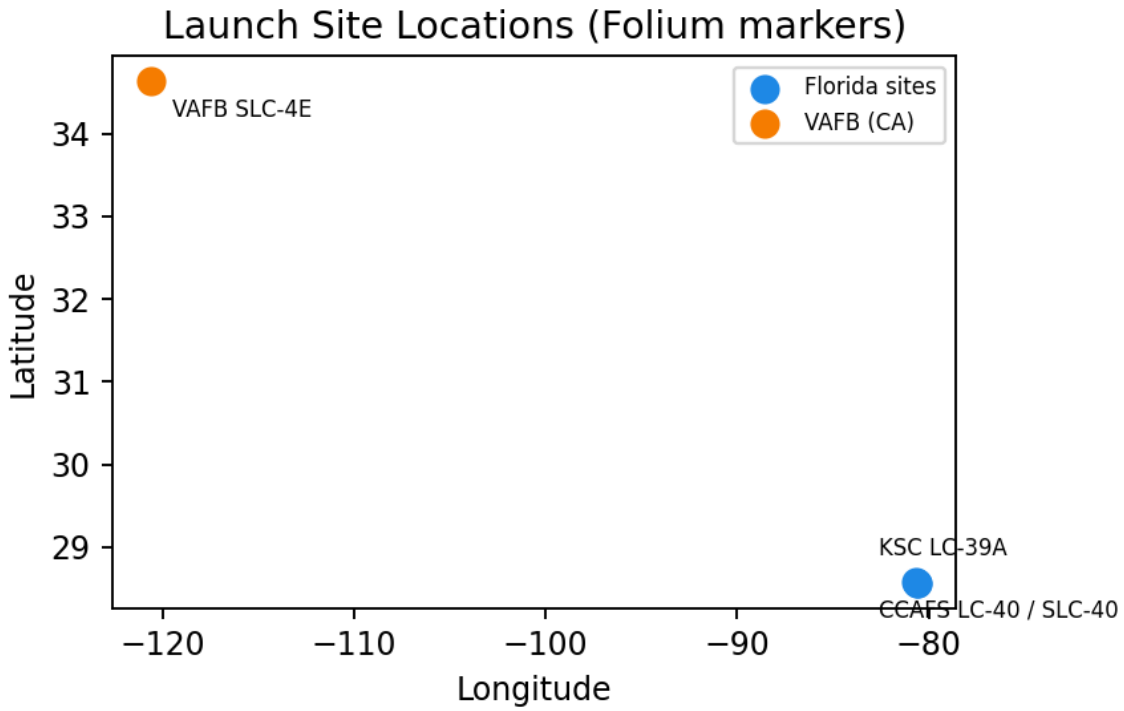
Query results from the SQL notebook; replace with your own output values if they differ.

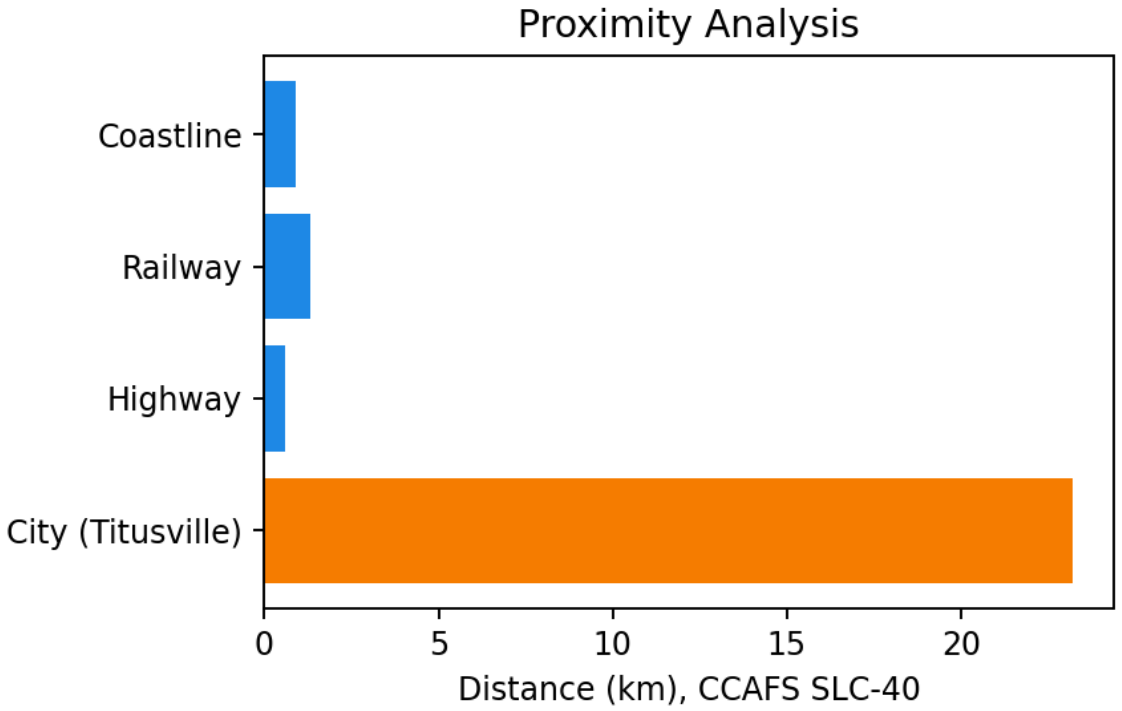
EDA with SQL Results: Rankings & Time Analysis

Landing outcome (2010-06-04 to 2017-03-20)	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- **Time analysis:** only two drone-ship failures in 2015 (Oct 10: B1012; Apr 14: B1015), both from CCAFS LC-40.
- Ranking shows early flights dominated by failures and no-attempt landings.

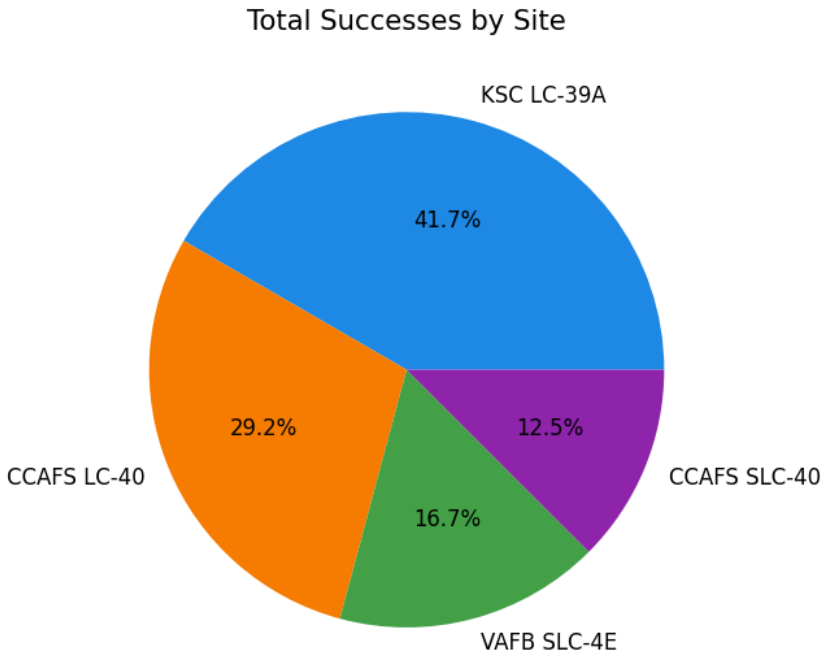
Folium Map: Markers, Launch Records & Proximity

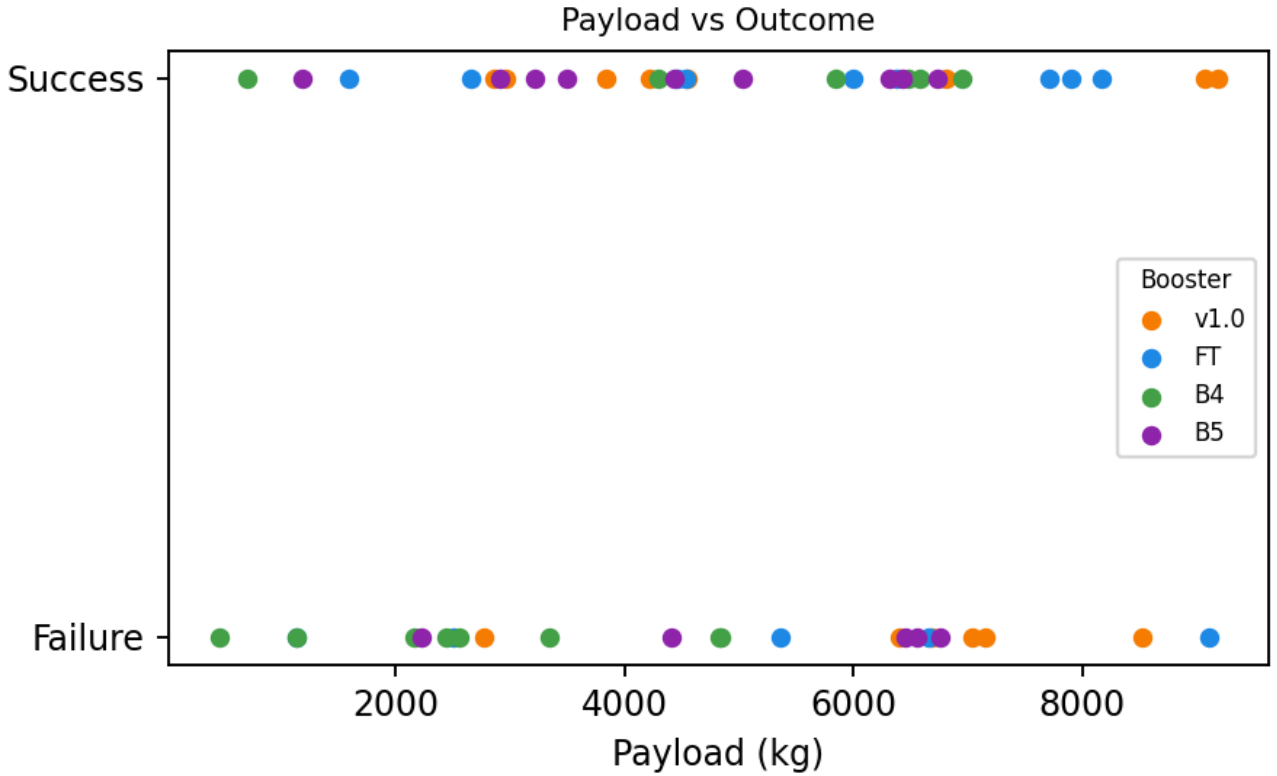




Green/red markers show success/failure per site. Sites sit close to the coastline (~0.9 km) for safe over-water launches, near railways/highways (~1–1.5 km) for logistics, and far from cities (>20 km).

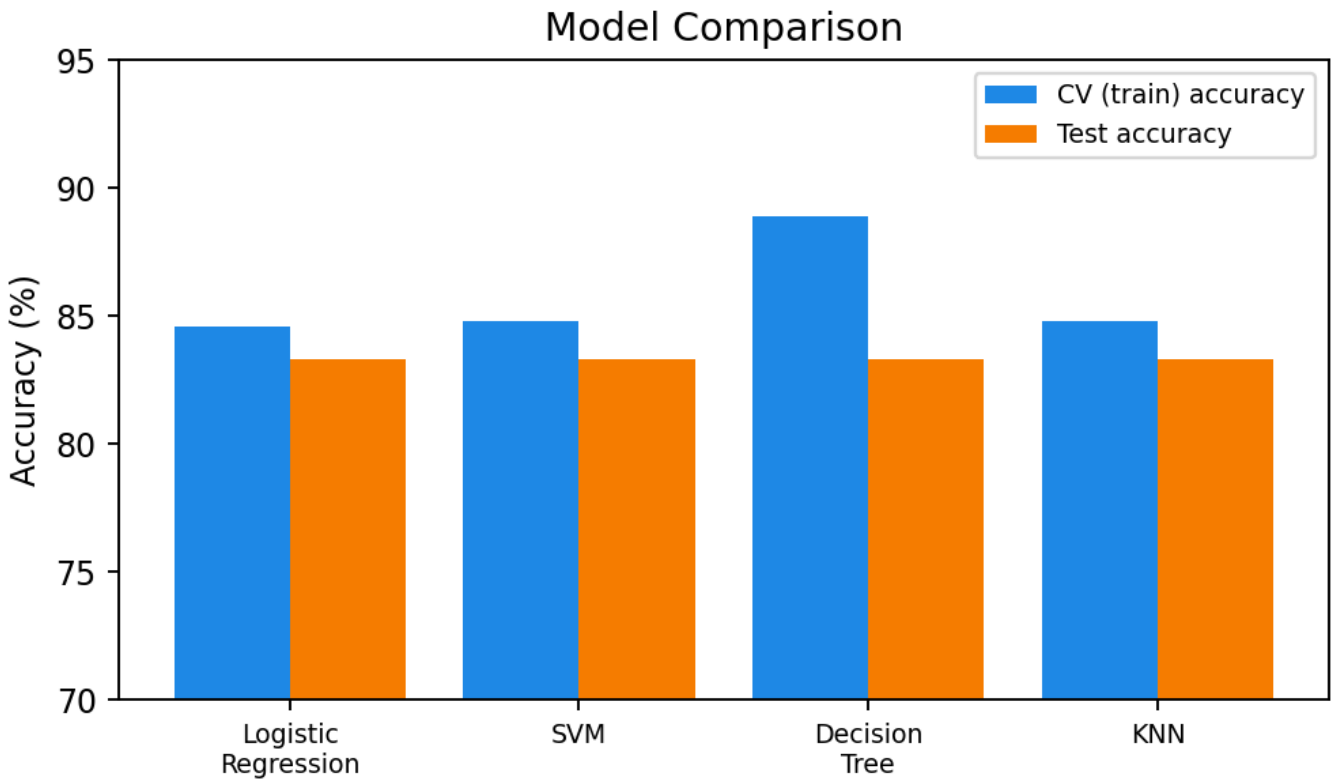
Plotly Dash: Success Pie & Payload vs Outcome





KSC LC-39A has the highest success ratio (76.9% of its launches); payloads 2,000–5,500 kg have the highest success; booster FT leads.

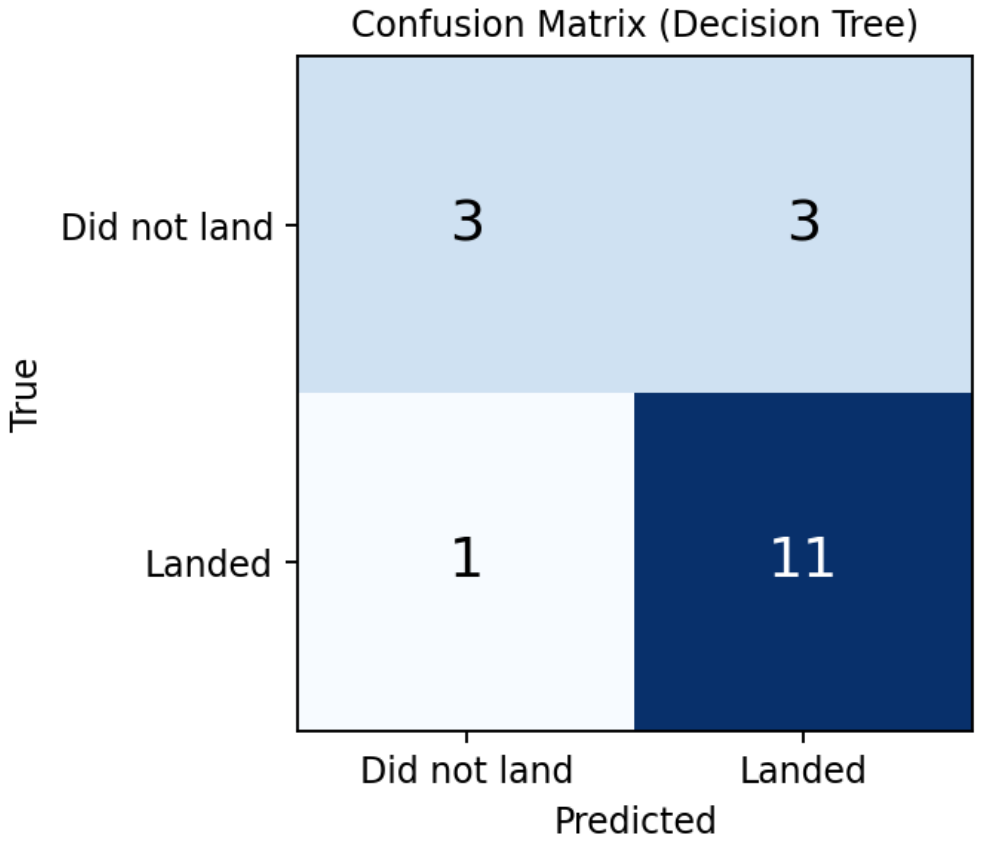
Predictive Analysis: Models & Evaluation



- **Process:** standardize features, 80/20 train-test split (18 test samples), GridSearchCV (cv=10) for each model.
- **Models:** Logistic Regression, SVM, Decision Tree, KNN.
- **Evaluation:** accuracy on test set is identical (83.3%) across models; cross-validation separates them.
- **Best:** Decision Tree (CV 88.9%).

> Small test set (18 samples) limits how much models can be separated; more data would give a clearer winner.

Predictive Analysis: Confusion Matrix & Best Model



- **Best model: Decision Tree** — highest cross-validated accuracy (88.9%), interpretable rules, handles mixed features.
- Confusion matrix: 11 true positives, 3 true negatives, 3 false positives, 1 false negative.
- Precision 0.79, recall 0.92, F1 0.85 for the 'landed' class.
- Main weakness: false positives (predicting a landing that fails).

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Conclusions & Innovative Insights

- Landing success improved dramatically from 2013 to 2020 as SpaceX gained experience (flight number is the strongest signal).
- KSC LC-39A and orbits ES-L1, GEO, HEO and SSO show the highest success rates; heavy-payload GTO missions are harder.
- Reused boosters with grid fins and landing legs (Block 5) land far more reliably.
- Decision Tree is the best predictor; models can estimate launch cost for competitor bids.
- **Insight:** combine a landing-probability score with launch cost to price bids; next steps: add weather, booster age, and gradient-boosted models; apply on newer launches.

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